THE UNIVERSITY OF AZAD JAMMU AND KASHMIR, MUZAFFARABAD

Department of Software Engineering

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LAB:	**04**
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COURSE CODE:	**CS-2203**
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Lab Report: Interfaces and Cabling

Objective: To understand the Physical Layer (Layer 1) of the OSI model by terminating and testing UTP cables according to industry standards.

1. BACKGROUND

The Physical layer governs the transmission of raw data over a medium. This lab focused on copper cabling using RJ45 connectors and Unshielded Twisted Pair (UTP) cable.

- **UTP (UNSHIELDED TWISTED PAIR):** Standard Ethernet cable for offices and homes. It relies on wire twisting to cancel electromagnetic interference (EMI). No external shielding is used.
- **STP (SHIELDED TWISTED PAIR):** Includes a metallic foil or braid around pairs to protect against high interference, typically used in industrial environments.

Cables are categorized by performance standards (e.g., Cat5e, Cat6) as defined by IEEE, with higher categories supporting faster data rates.

2. WIRING STANDARDS (EIA/TIA)

To ensure interoperability, two standard pinouts exist for the 8-pin RJ45 connector. This lab utilized the widely adopted **T568B** standard.

Pin	T568B Color	Function
1	White-Orange	Transmit +
2	Orange	Transmit -
3	White-Green	Receive +
4	Blue	Unused

5	White-Blue	Unused
6	Green	Receive -
7	White-Brown	Unused
8	Brown	Unused

3. CABLE CONFIGURATIONS

Depending on the devices being connected, different wiring schemes are used:

- **STRAIGHT-THROUGH CABLE:** Both ends wired with T568B. Used to connect different devices (e.g., PC to Switch, Router to Switch).
- **CROSSOVER CABLE:** One end T568A, the other T568B. Used to connect similar devices (e.g., PC to PC, Switch to Switch). *Note: Modern devices often support Auto MDI-X, making crossover cables less critical.*
- ROLLOVER CABLE: Pin 1 connects to Pin 8, Pin 2 to Pin 7, etc. Used specifically for console connections to manage routers and switches.

4. TOOLS USED

The following tools were essential for cable fabrication and testing:

- **RJ45 CONNECTORS (8P8C):** The plastic plugs terminating the cable ends.



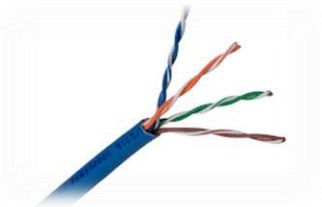
- **CRIMPING TOOL:** A multi-purpose tool for cutting, stripping, and crimping connectors onto the cable.



- **WIRE STRIPPER/CUTTER:** Used to remove the outer jacket without damaging internal wires.



- **UTP CABLE (CAT5E/6):** Bulk cable medium for creating custom-length patch cords.



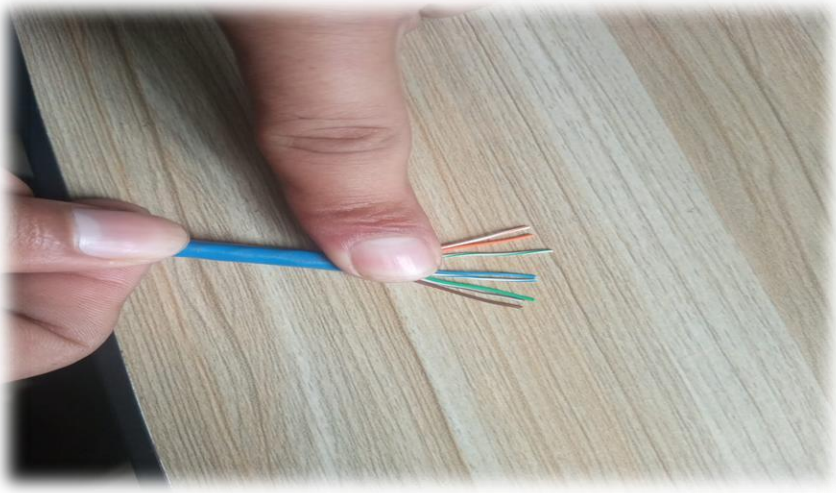
- **CABLE TESTER:** Verifies continuity and correct wire mapping (pins 1 through 8).



5. PROCEDURE

PREPARE: 1. Cut a suitable length of UTP cable and strip 1-2 inches of the outer jacket to expose the four twisted pairs.

2. **ARRANGE:** Untwist the pairs and arrange them in the **T568B** order: *White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown.*



3. **TRIM & INSERT:** Trim the wires evenly and insert them fully into the RJ45 connector, ensuring they reach the end.



4. **CRIMP:** Firmly compress the crimping tool to secure the connector.
5. **TEST:** Connect both ends of the cable to the cable tester and observe the LED sequence.

6. RESULTS

The cable tester confirmed a successful build. All eight LEDs illuminated sequentially (1 through 8) on both the local and remote units, indicating:

- **CONTINUITY:** No breaks or "opens" in the wires.
- **CORRECT WIRE MAP:** No wires were crossed or shorted.
- **STANDARD COMPLIANCE:** The cable functions as a Straight-Through T568B cable.



7. CONCLUSION

This lab successfully demonstrated physical layer implementation. By fabricating a working straight-through cable according to the T568B standard and validating it with a tester, we established foundational skills in media termination and network troubleshooting.